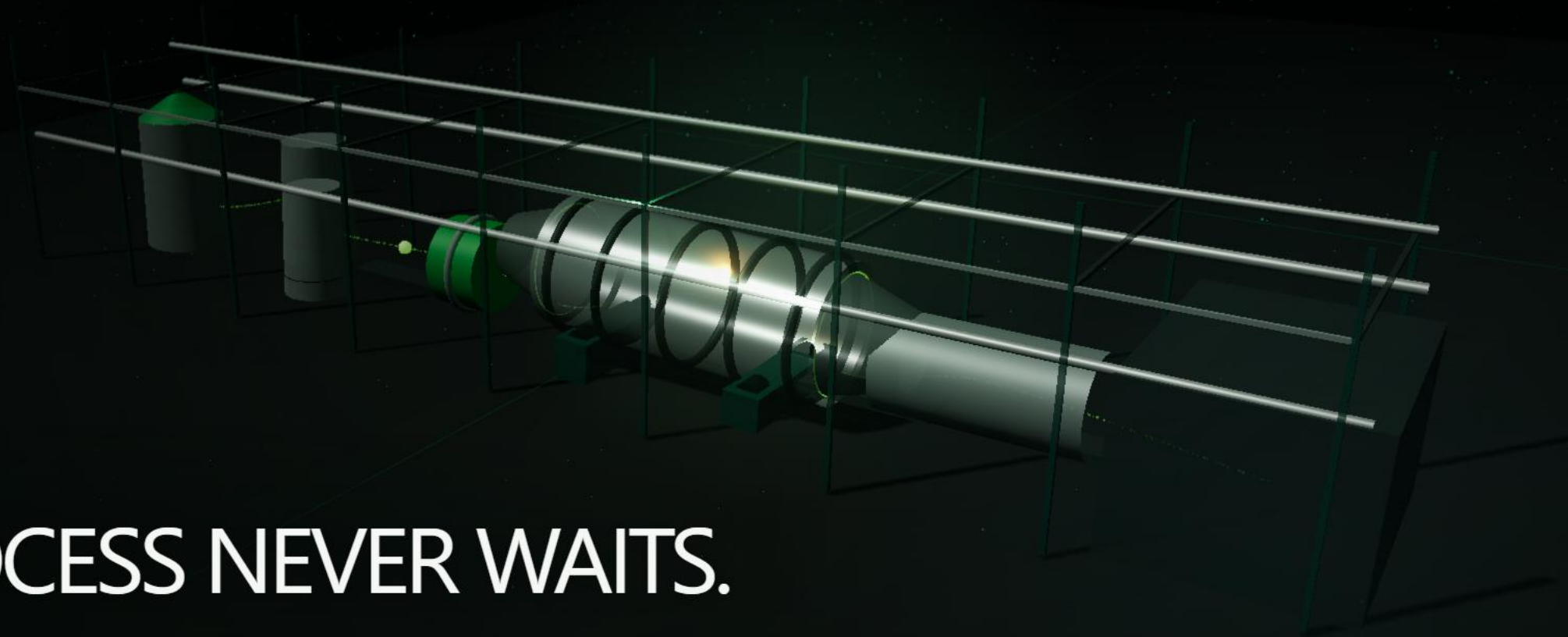




OCP / SOLUBLE MAP

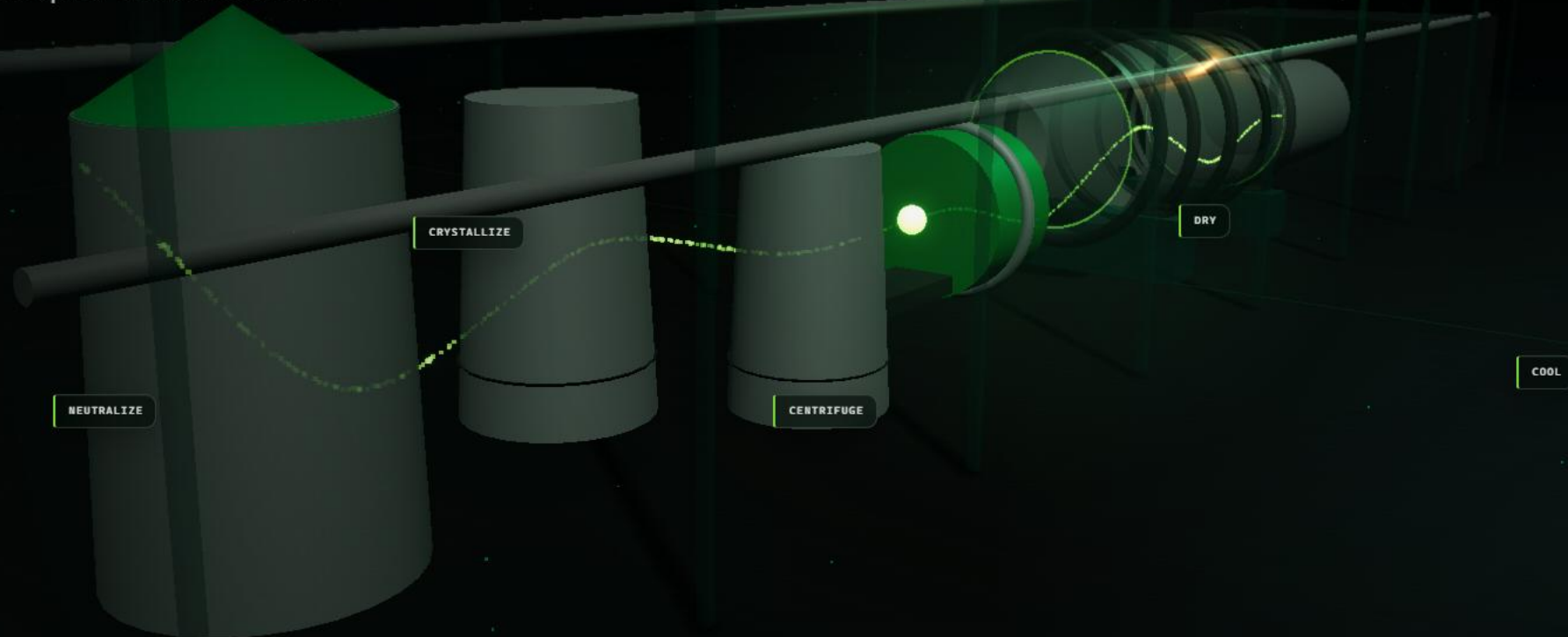
THE PROCESS NEVER WAITS.

So visibility cannot either.

01 / PROCESS JOURNEY

FOLLOW THE GRANULE.

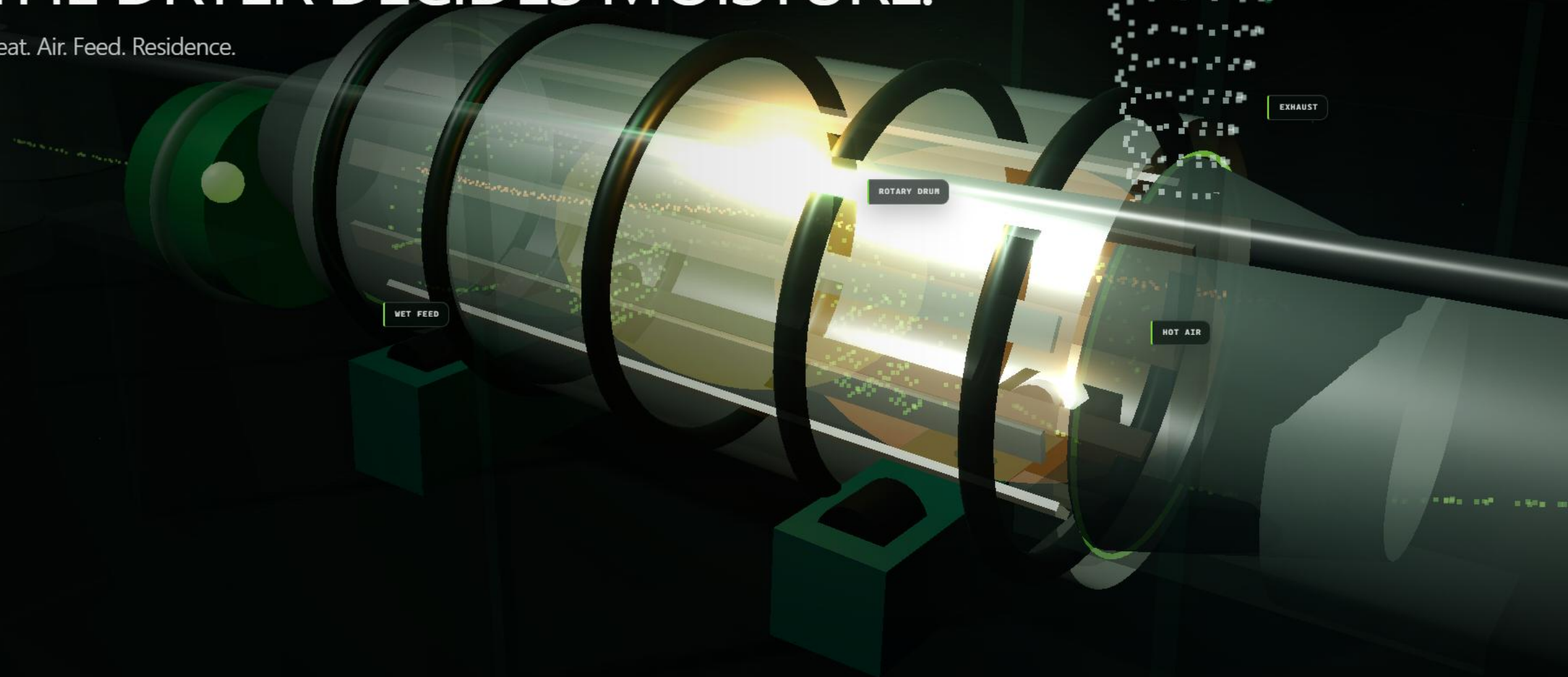
One product. Five coupled transformations.



02 / HERO ASSET

THE DRYER DECIDES MOISTURE.

Heat. Air. Feed. Residence.





MOISTURE LEAVES IN MOTION.

Rotation. Lift. Cascade. Evaporation.

SCHEMATIC MECHANISM

04 / THE VISIBILITY GAP

TWO HOURS OF SILENCE.

The process keeps writing its story.

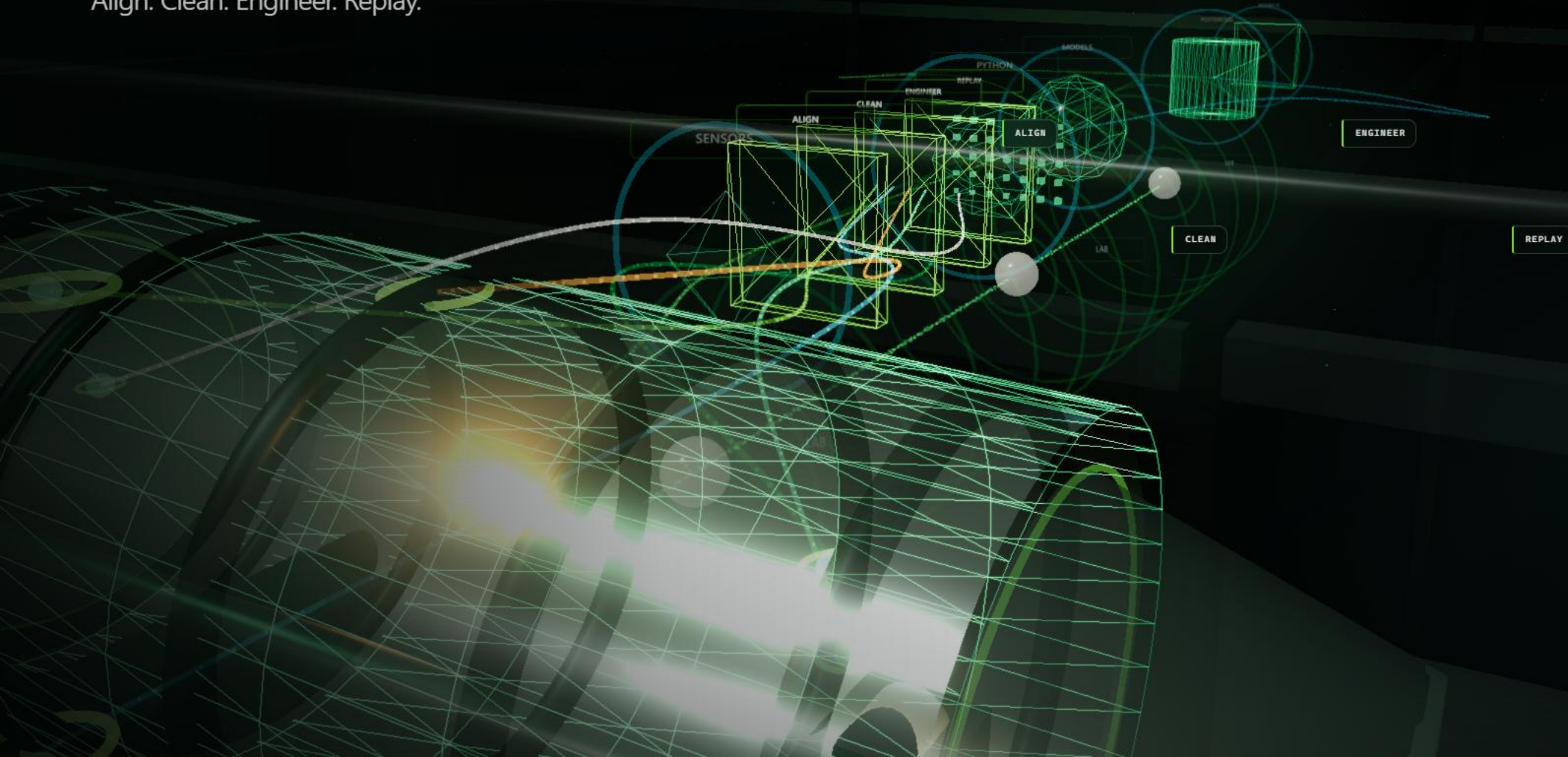
LAB ~ 2 h

REPLAY $dt = 5\text{ s}$ 

05 / PREPROCESSING

SIGNALS BECOME STRUCTURE.

Align. Clean. Engineer. Replay.



06 / END-TO-END ARCHITECTURE

ONE PIPELINE. FIVE LAYERS.

Sensors -> Python -> models -> PostgreSQL -> Power BI



07 / MODEL INTELLIGENCE

TWO QUESTIONS. TWO BRANCHES.

US ADVISORY ESTIMATE

LEARNED NORMAL ENVELOPE

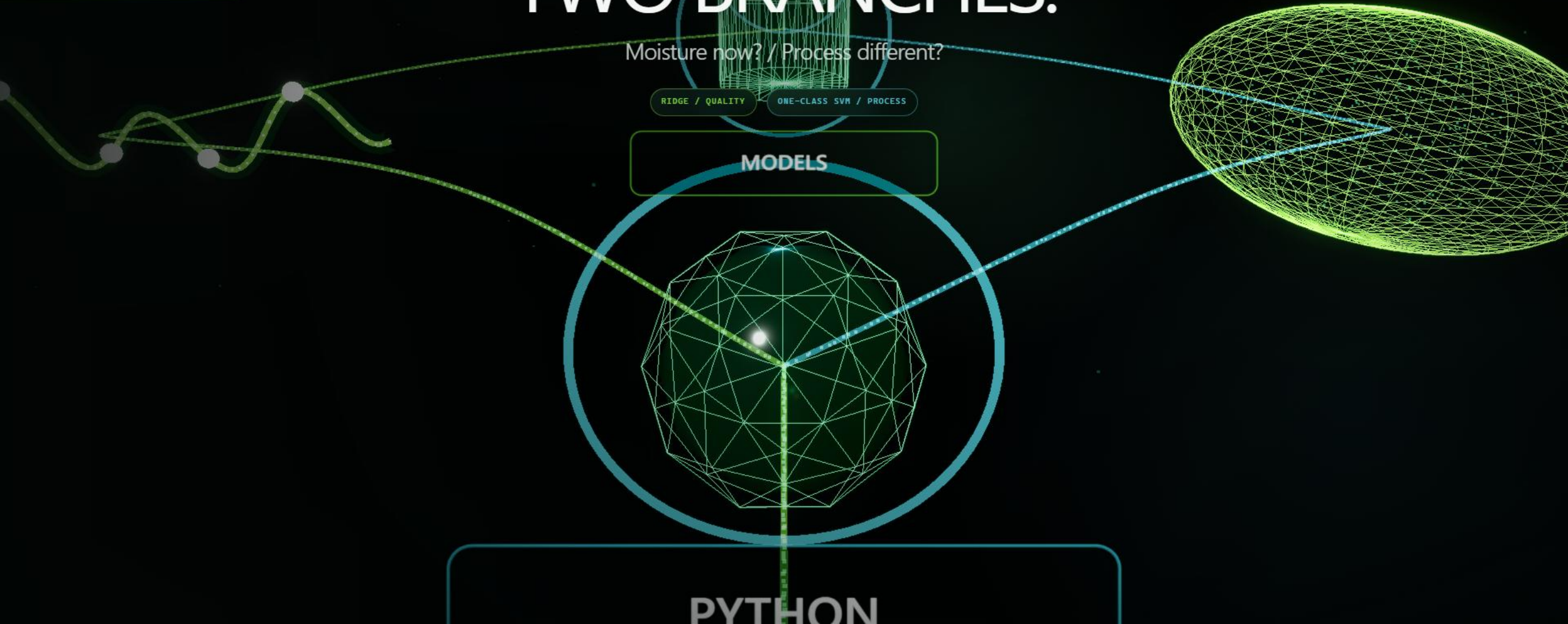
Moisture now? / Process different?

RIDGE / QUALITY

ONE-CLASS SVM / PROCESS

MODELS

PYTHON



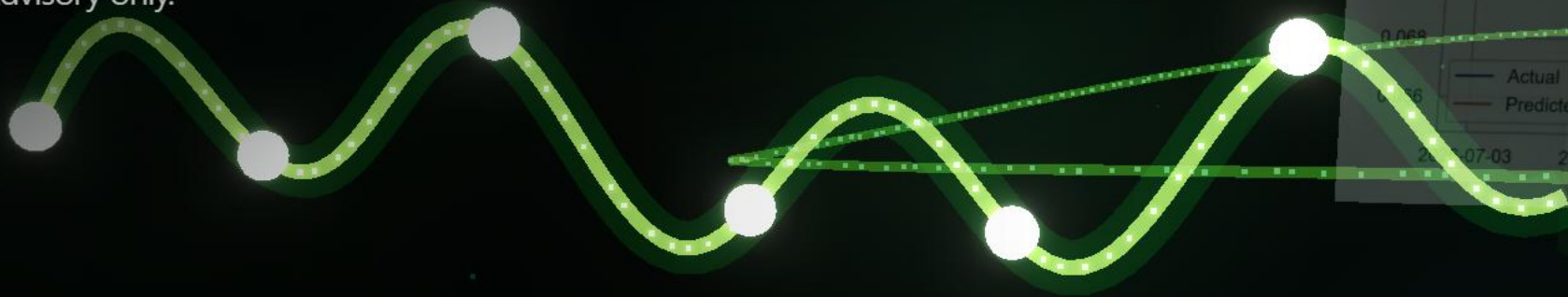
08 / QUALITY BRANCH

RIDGE ESTIMATES BETWEEN LABS.

alpha = 10. 16 features. Advisory only.

LAB REMAINS THE REFERENCE

CONTINUOUS ADVISORY ESTIMATE



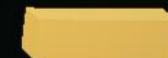


THE SVM RECOGNIZES NOVELTY.

15 process-only features. Evidence, not causality.

5.925% validation flags

RISK != PROBABILITY



10 / HELD-OUT EVIDENCE

EVIDENCE BEFORE AMBITION.

Synthetic prototype. Chronological test. Advisory only.

R2

0.8245

MAE

0.001069

RMSE

0.001403

TEST / 165 LAB SAMPLES / ERROR UNITS: PERCENTAGE POINTS % H2O

— Actual
— Predicted

2026-07-03

2026-07-05

2026-07-07

2026-07-09

2026-07-11

2026-07-13

2026-07-15

2026-07-17

Timestamp



11 / OPERATOR SUPERVISION

PREDICTED FINAL MOISTURE

0.80 %

LATEST LAB MOISTURE

0.75 %

Sampled 10:00 · 1 min ago

VALIDATED MOISTURE ERROR

0.060 %

Validated at 10:00

CURRENT ANOMALY RISK

0.10

Warning 0.50 · Critical 0.80

PROCESS STATUS

NORMAL

Stable operation

ANOMALY SEVERITY

NORMAL

Normal operation

SUPERVISION, NOT AUTOMATION.

Power BI visualizes. Python runs models.

AUTHENTIC PROJECT PREVIEWS

NO MODEL EXECUTION IN POWER BI

POWER BI

Current diagnosis context

Latest explainable inference and validation context

Subsystem

Normal operation

Diagnosis

No active anomaly

Last event

2026-08-06 09:24:20

Model

Ss_v1.0

Critical process variables

Latest values and rolling 8-hour trends; color = state

Dryer air temp.

177.1 °C

Δ +1.0

STALE

Wet feed rate

25.9 t/h

Δ -0.2

STALE

Steam pressure

8.24 bar

Δ +0.0

STALE

Air flow rate

24,083 m³/h

Δ -54.8

STALE

Vacuum

47.9 mmH2O

Δ -0.7

STALE

Operator guidance

Explainable diagnosis and recommended checks — advisory only

Possible causes

No anomaly-related cause list is generated.

Recommended verification

Continue normal monitoring and verify data freshness and instrument quality.

Decision support only — model diagnostics are advisory evidence, not confirmed root cause and not automatic process control. Rendered from live SQL views 2026-06-14 13:48:04.

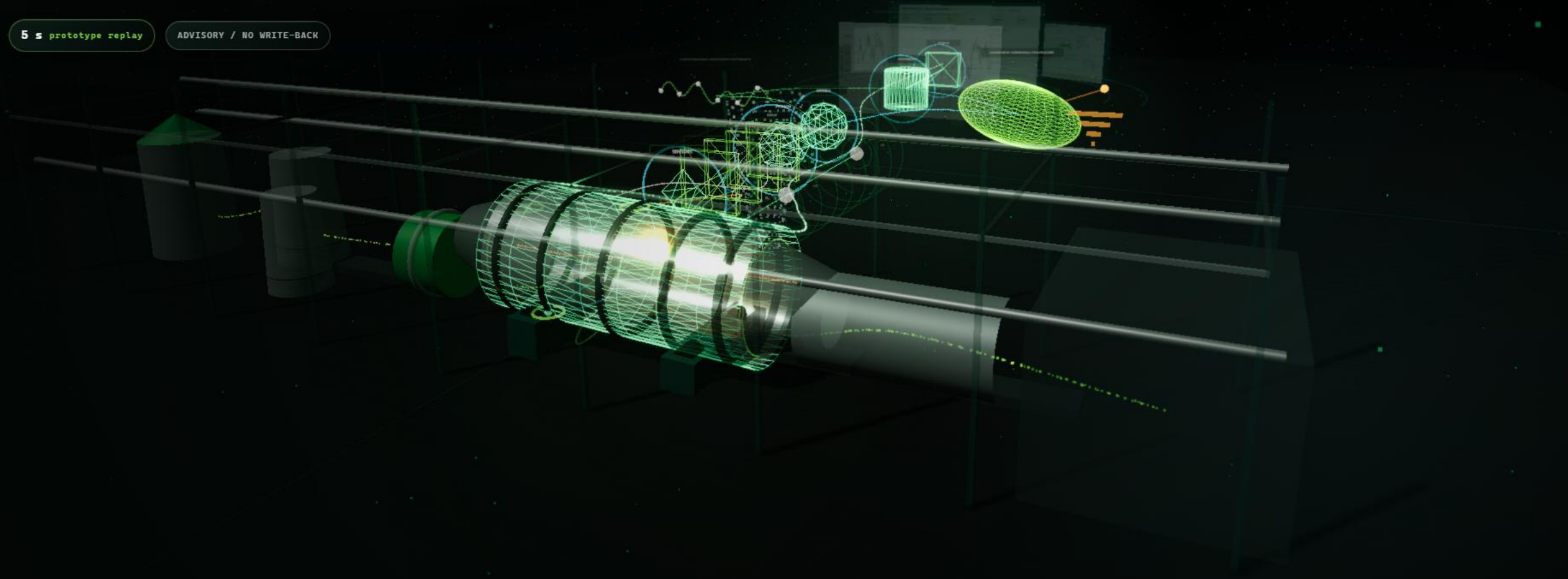
12 / SYSTEM IN MOTION

ONE OPERATING INTELLIGENCE LOOP.

Replay -> predict -> detect -> publish -> supervise

5 s prototype replay

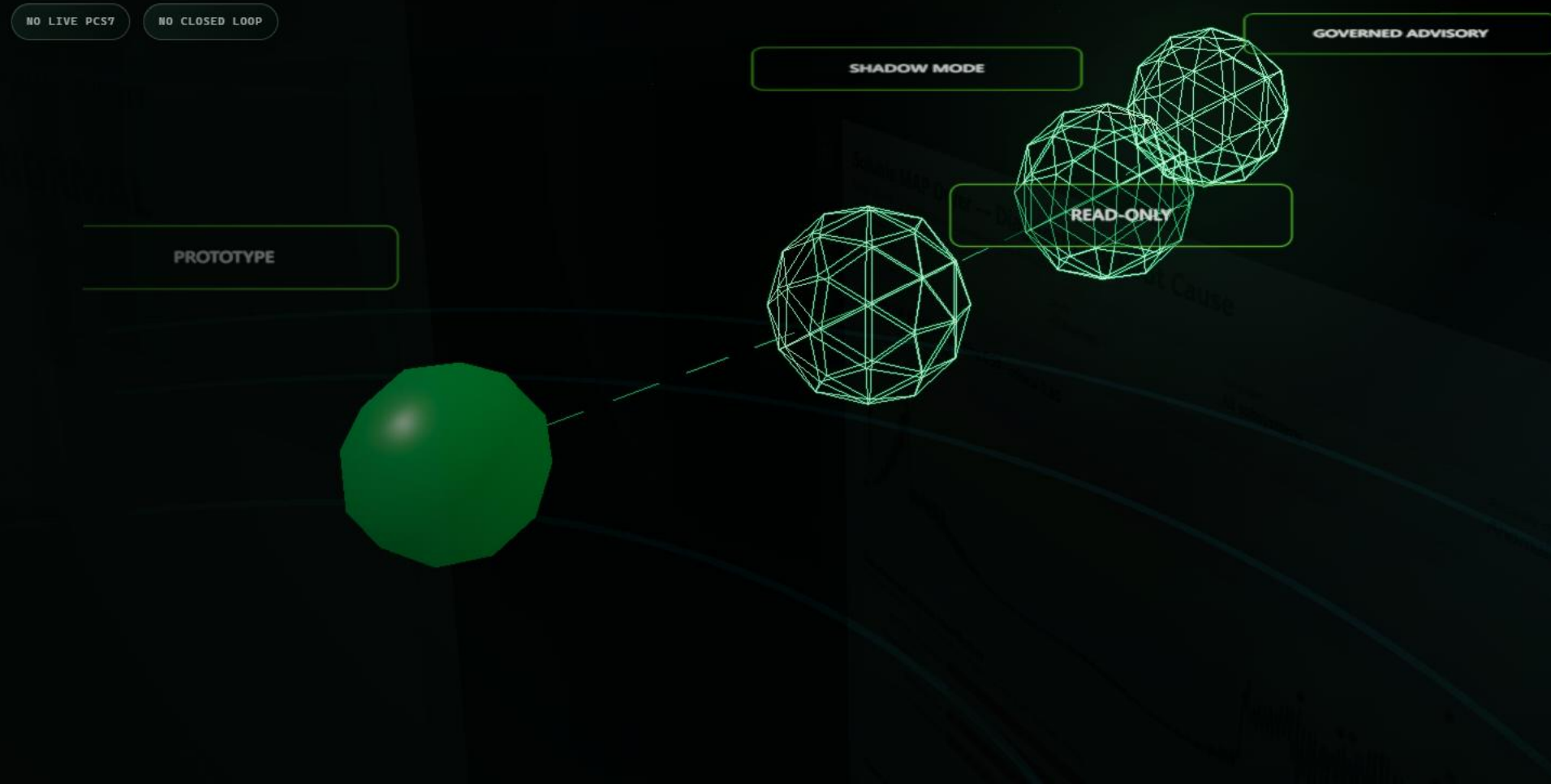
ADVISORY / NO WRITE-BACK



13 / GOVERNED SCALE-UP

PROVE IT IN SHADOW MODE.

Read-only plant data -> validation -> governed advisory



SOLUBLE MAP / DRYER DIGITALIZATION

THE DRYER KEEPS MOVING.

Digital visibility continues beyond laboratory delay.

LABORATORY MOISTURE REMAINS THE REFERENCE